

Rohit Yerrarapu

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Career Objective

Aspiring AI Engineer and Computer Science undergraduate with a strong interest in Artificial Intelligence, Machine Learning, Speech Processing, and Full-Stack Development. Experienced in building AI-powered applications, multimodal deep learning systems, educational platforms, and intelligent web solutions. Passionate about developing impactful AI-driven products and scalable software systems.

Education

Hyderabad Institute of Technology And Management 2023 – Present
Bachelor of Technology (B.Tech), Computer Science and Engineering

Sri Chaitanya Junior College 2021 – 2023 Senior Secondary Education
Sri Chaitanya Techno Schools 2018 – 2021 Secondary Education

Technical Skills

Programming Languages: Python, Java, JavaScript, TypeScript, SQL

AI Machine Learning: PyTorch, Hugging Face Transformers, BERT, HuBERT, Scikit-learn, NumPy, Pandas, Deep Learning, NLP, Speech Processing

Web Development: React.js, Next.js, HTML, CSS, Tailwind CSS, Streamlit, Flask, Django, FastAPI

Databases Cloud: Supabase, PostgreSQL, SQLite, Firebase

Tools: Git, GitHub, VS Code, Jupyter Notebook

Work Experience

Software & AI Development Intern Feb 2026 – Present
KODESPARK

- Led frontend development for a Movie Review Platform using React.js and modern web technologies.
- Developed responsive and interactive user interfaces focused on user experience and performance.
- Worked on AI-powered application features and software development workflows.
- Contributed to microservices-based application architecture and system integration.
- Collaborated with development teams on feature implementation, testing, and deployment.

Software Development Intern Sep 2025 – Dec 2025
GNU Hive (OPC) Private Limited, Remote

- Worked on software development and web application projects under industry mentorship.
- Participated in debugging, testing, and feature implementation activities.
- Collaborated using Git and modern software development workflows.
- Strengthened analytical thinking and problem-solving skills.

AI Developer Intern – VIRASAM Jun 2025 – Jul 2025
AI-Powered Cultural Heritage Preservation Platform

- Developed the complete application architecture for an AI-powered cultural heritage platform.
- Integrated Gemini AI models using Genkit.
- Built responsive user interfaces using Streamlit.
- Implemented authentication and database management using Supabase.
- Optimized application workflows and user experience.

Projects

Multimodal Emotion Recogniser

- Developed a multimodal deep learning framework for emotion recognition using speech and text.
- Built a CNN + BiLSTM speech classifier and BERT-based text emotion classifier.
- Designed a Late Fusion Neural Network achieving 98.9% classification accuracy.
- Reduced inference latency from 2000ms to 180–240ms using optimization techniques.
- Developed a real-time dashboard with spectrogram visualization and live model evaluation.

Native Language Identification Using HuBERT

- Processed and analyzed 8,116 speech recordings from the IndicAccentDB dataset.
- Extracted MFCC features and HuBERT embeddings for language classification.
- Achieved 100% accuracy using MFCC features and 98
- Built machine learning models to identify native language backgrounds from English speech.

EduPredict – AI-Powered Academic and Classroom Management System

- Developed a full-stack academic management platform with AI-powered insights.
- Integrated Gemini AI using Genkit and Supabase.
- Built attendance tracking, performance analytics, and academic monitoring dashboards.
- Implemented role-based workflows for students, faculty, and administrators.

Mockr – AI-Powered Interview Preparation Platform

- Developed an AI-powered interview preparation platform for technical and HR interviews.
- Implemented automated feedback generation and performance analytics.
- Built personalized interview practice workflows using AI-generated content.

Achievements

- Built a multimodal emotion recognition system achieving 98.9
- Processed and analyzed over 8,000 speech samples for language identification research.
- Developed multiple AI-powered applications integrating Gemini AI, Genkit, and Supabase.
- Optimized deep learning inference pipelines reducing latency from 2000ms to 180–240ms.
- Successfully completed software development and AI-focused internships.